Axler Linear Algebra

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Contents

I plan to mostly take notes on paper, but I will a lot of the exercises here because I need to practice writing LaTeX quickly and stuff.

1 Vector Spaces

1.A $\mathbb{R}^n$ and $\mathbb{C}^n$

These exercises are really easy but I am always scared that my proofs of these are not rigorous enough....

I do odds.

1 Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Proof. Let $\alpha = a + bi$ and $\beta = c + di$ where $a, b, c, d \in \mathbb{R}$.

Then we have

$$\begin{aligned} \alpha + \beta = \\ a + bi + c + di = \\ (a + c) + (b + d)i = \end{aligned}$$

Because addition is commutative under the reals, we can swap the terms.

$$\begin{aligned} (c + a) + (d + b)i = \\ c + di + a + bi = \\ \beta + \alpha \end{aligned}$$

□

3 Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Distribute, multiply, then rearrange terms using the communitivity of the reals to get the answer. It is quite tedious, so I won't do it here.

5 Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Proof. Let $\alpha = a + bi$. Then, let $\beta = -a - bi$. We can observe that

$$\begin{aligned} \alpha + \beta = \\ a + bi - a - bi = \\ (a - a) + i(b - b) = 0 \end{aligned}$$

□

7 Show that $\frac{-1+\sqrt{3}i}{2}$ is a cube root of 1 (meaning that its cube equals 1)

Again, this is a tedious problem where you just multiply it out. Of course, I can represent this number as $e^{\frac{\pi}{3}i}$ but that's cheating.

9 Find $x \in \mathbf{R}^4$ such that
$(4, -3, 1, 7) + 2x = (5, 9, -6, 8)$

Proof. This is vector addition, so we can apply the definition.

$$\begin{aligned} (4, -3, 1, 7) + 2x &= (5, 9, -6, 8) \\ (4 + 2x_1, -3 + 2x_2, 1 + 2x_3, 7 + 2x_4) &= (5, 0, -6, 8) \end{aligned}$$

For each term in the list, the number must hold, so in essence we get four different equations:

$$\begin{aligned} 4 + 2x_1 &= 5 \\ -3 + 2x_2 &= 0 \\ 1 + 2x_3 &= -6 \\ 7 + 2x_4 &= 8 \end{aligned}$$

Each of these linear equations yields only one solution, so the only solution for the equation is $\boxed{x = (\frac{1}{2}, -\frac{3}{2}, -\frac{7}{2}, \frac{1}{2})}$ □

1.B Definition of a Vector Space

This one introduces the concept of a vector space in an abstract manner and gives a few unorthodox examples.

1 Prove that $-(-v) = v$ for every $v \in V$

Proof. $-(-v)$ is the additive inverse of $-v$. Therefore,

$$\begin{aligned} -(-v) + (-v) &= 0 \\ -(-v) + (-v + v) &= v \\ -(-v) + 0 &= v \\ -(-v) &= v \end{aligned}$$

As desired. □

3 Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

$$\begin{aligned} v + 3x &= w \\ 3x &= w - v \\ x &= \frac{1}{3}(w - v) \end{aligned}$$

Since that right side is a unique vector, x is a unique vector. You invoke the properties and stuff for this, but I'm too lazy.

5 Show that in the definition of a vector space (1.20), the additive inverse condition can be replaced with the condition that

$$0v = 0 \text{ for all } v \in V.$$

Here the 0 on the left side is the number 0, and the 0 on the right side is the additive identity of V.

Proof. We can show that given this condition, every element $v \in V$ has an additive inverse.

$$\begin{aligned} 0v &= 0 \\ (1 + -1)v &= 0 \end{aligned}$$

Note that even though we don't know whether *vector* addition has an additive inverse the 0 on the left size is just a scalar, so we can still say there is an additive inverse.

Then distribute and apply the multiplicative identity axiom...

$$\begin{aligned} 1v + (-1)v &= 0 \\ v + (-1)v &= 0 \end{aligned}$$

Therefore $(-1)v$ is the additive inverse for all $v \in V$. □

7 Suppose S is a nonempty set. Let V^S denote the set of functions from S to V. Define a natural addition and scalar multiplication on V^S , and show that V^S is a vector space with these definitions.

Proof. Like with F^S, we can sort of "piggyback" off the properties of V to maintain the properties of a vector space in V^S.

Define addition as

$$(f + g)(x) = f(x) + g(x)$$

Let the additive identity be $0(x) = 0$ for all $x \in S$.

$$(f + 0)(x) = f(x) + 0(x) = f(x) + 0 = f(x)$$

For all $f, g \in V^S$. Because $f(x)$ is a vector, it has an inverse (namely $-f(x)$). Therefore, we can define the additive inverse as

$$(-f)(x) = -f(x)$$

$$(f + (-f))(x) = f(x) + (-f(x)) = 0 = 0(x)$$

Communativity:

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$$

We can swap $f(x)$ and $g(x)$ because V is communative.

Associativity:

$$((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x)$$

Again, we use the properties of the vector space V to help us out here.

Scalar multiplication

For all $\lambda \in \mathbb{F}, v \in V^S$,

$$(\lambda v)(x) = \lambda v(x)$$

Multiplicative Identity

1 is the multiplicative identity.

$$(1v)(x) = 1v(x) = v(x)$$

Since 1 is the multiplicative identity of V.

Distributive property

$$(a(u + v))(x) = a(u + v)(x) = a(u(x) + v(x)) = au(x) + av(x) = (au + av)(x)$$

Because of the distributive property of V.

$$((a + b)v)(x) = (a + b)v(x) = av(x) + bv(x) = (av + bv)(x)$$

For the same reason.

Closure

Because V is closed under addition and scalar multiplication, and all our operations are defined using these 2 operations, V^S is closed.

Thus, we have proved all the properties of vector spaces on V^S, therefore V^S is a vector space.

$\square$

1.C Subspaces

This is where it ramps up a little bit, we are doing some linear algebra now! This idea of "subspaces" is interesting...

1 For each of the following subsets of $\mathbf{F}^3$, determine whether it is a subspace of $\mathbf{F}^3$

(a) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$

Proof. $(0, 0, 0)$ is in our set, so we have the additive identity. Consider vectors x and y in our set, and add them.

$$\begin{aligned} x + y &= \\ (x_1, x_2, x_3) + (y_1, y_2, y_3) &= \\ (x_1 + y_1, x_2 + y_2, x_3 + y_3) &= \\ z &= \\ z_1 + 2z_2 + 3z_3 &= \\ x_1 + y_1 + 2x_2 + 2y_2 + 3x_3 + 3y_3 &= \\ x_1 + 2x_2 + 3x_3 + y_1 + 2y_2 + 3y_3 &= \\ 0 + 0 &= 0 \end{aligned}$$

Because x and y are members of the subset, we can use the condition. Therefore, z is part of our set.

Now let us consider scalar multiplication. Consider x in our set and $\lambda \in \mathbb{F}$. Then,

$$\begin{aligned} \lambda x &= \\ (\lambda x_1, \lambda x_2, \lambda x_3) \end{aligned}$$

We then consider whether this new element is in our set.

$$\begin{aligned} \lambda x_1 + 2\lambda x_2 + 3\lambda x_3 &= \\ \lambda(x_1 + 2x_2 + 3x_3) &= \\ \lambda(0) &= 0 \end{aligned}$$

We use the distributive property here. Since it follows the set's condition, we can say that it is closed under scalar multiplication.

These are all the conditions, so we can say that our set is a subspace of $\mathbb{F}^3$. $\square$

(b) $\{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$

Proof. Consider the element $(1, 0, 1)$ which is in our set because $1+2\cdot 0+3\cdot 1 = 4$ and it is in $\mathbb{F}^3$. Now consider multiplying it by the scalar 2 to create the element $(2, 0, 2)$. Unfortunately, $2 + 2 \cdot 0 + 3 \cdot 2 = 8 \neq 4$, so this set does not fulfill the closure under scalar multiplication condition. $\square$

(c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1x_2x_3 = 0\}$

Proof. Consider the sets $(1, 1, 0)$ and $(0, 0, 1)$. Summing these up, we get $(1, 1, 1)$, but $1 \cdot 1 \cdot 1 = 1 \neq 0$, so this set fails the additive closure property. ☐

(d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$
I am too lazy but I did the other 3, good enough

2 I did number 2 in my notebook, not noticing it was actually an exercise.

3 Show that the set of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbf{R}^{(-4,4)}$.

Proof. Clearly this set is a subset of $\mathbb{R}^{(-4,4)}$. The zero function ($f(x) = 0$) satisfies the condition (since the derivative is also 0), so we have the additive identity.

Consider functions f, g in the set, then consider $(f + g)(x)$. Functions and their derivatives are additive, so we can write

$$\begin{aligned} (f+g)'(-1) &= \\ f'(-1) + g'(-1) &= \\ 3f(2) + 3g(2) &= \\ 3(f(2) + g(2)) &= \\ 3(f+g)(2) \end{aligned}$$

Therefore, we have additive closure.

Consider f in the set and $\lambda \in \mathbb{R}$. Let us show that $(\lambda f)(x) = \lambda f(x)$ is in the set.

$$\begin{aligned} (\lambda f)'(-1) &= \\ \lambda f'(-1) &= \\ 3\lambda f(2) &= \\ 3(\lambda f)(2) \end{aligned}$$

So we are closed under scalar multiplication. ☐

5 Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Proof. Although $\mathbb{R}^2$ is a vector space over $\mathbb{R}$, it is not a vector space over $\mathbb{C}$. For example, multiplying $(1, 1)$ by i produces a vector not in $\mathbb{R}^2$, so it is not closed under scalar multiplication. ☐

7 If U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbb{R}^2$.

Proof. Consider the set $\mathbb{Q}^2$. It is closed under addition and additive inverse, but it is *not* closed under scalar multiplication over $\mathbb{R}$. For example, the set $(1, 1)$ when multiplied by an irrational number such as $\sqrt{2}$ is not a part of the set. ☐

9 A function $f : \mathbb{R} \to \mathbb{R}$ is called periodic if there exists a positive number p such that $f(x) = f(x+p)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$? Explain.

Proof. There is not closure under addition, unfortunately. Consider the functions

$$\begin{aligned} f(x) &= \cos(2\pi x) \\ g(x) &= \cos(2x) \end{aligned}$$

$f(x)$ is at its maximum of 1 whenever $x \in \mathbb{Z}$, and $g(x)$ is at its maximum of 1 whenever $x = n\pi, n \in \mathbb{Z}$. $(f+g)(x) = 2$ only when both of these conditions are met. However, this is only true when $x = 0$.

Suppose that $x = n\pi$ and $x = m$ where $n, m \in \mathbb{Z}$. Then,

$$\begin{aligned} n\pi &= m \\ \pi &= \frac{m}{n} \end{aligned}$$

Which implies that π is rational if $n \neq 0$, so the only solution is $x = 0$.

Therefore, $(f+g)(x)$ reaches its maximum only at $x = 0$, and there is no $p > 0$ where $(f+g)(0) = (f+g)(p)$. Therefore $(f+g)(x)$ is outside the set and it fails the additive closure condition, making it not a vector space. □

11 Prove that the intersection of every collection of subspaces of V is a subspace of V.

Proof. Consider subspaces of V, $V_1, \dots, V_M$. All subspaces contain the identity element, so $V_1 \cap \dots \cap V_M$ will also contain it. Consider $u, v \in V_1 \cap \dots V_M$. Because $u, v \in V_k$, any addition and scalar multiplication will remain in V_k. Therefore, any addition and scalar multiplication will remain in $V_1 \cap \dots \cap V_M$, completing the proof. □

12 Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

First, lets prove a lemma we will need for this.

Lemma. *Given $u \notin V, v \in V, u + v \notin V$.*

Proof. Suppose that $u+v \in V$. Then, because of the additive inverse, $u+v-v = u \in V$, and we reach a contradiction. □

Now we can prove the main result.

Proof. Consider subspaces V_1, V_2

Consider WLOG $V_1 \subseteq V_2$. Then, $V_1 \cup V_2 = V_2$, and since V_2 is a subspace so is $V_1 \cup V_2$.

The converse is trickier. We prove the contrapositive. Consider when V_1 and V_2 are not subsets of each other. Then, there is some element $u \in V_1, u \notin V_2$. There must also be an element $v \in V_2, v \notin V_1$. These are true from the definition of subset. Consider $u + v$. By the lemma, this sum is in neither set, so it is not in the union of V_1 and V_2, so $V_1 \cup V_2$ is not a subspace.

□

13 Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

This problem is very difficult (as the textbook alluded) and I needed to look up the solution.

Proof. If any subspace is contained within another this reduces to the 2-subspace case.

Presume that $V_1 \cup V_2 \cup V_3$ is a subspace.

Suppose $V_1 \cup V_2 \backslash V_3$ was not empty. Then, we can use $u \in V_1 \cup V_2 \backslash V_3$ and $v \in V_3 \backslash V_2 \cup V_1$ to prove that there is not closure using the lemma from exercise 12. By symmetry, there is no element that is in exactly two of the subsets.

From here on, suppose $u \in V_1 \backslash V_2 \cup V_3$, $v \in V_2 \backslash V_1 \cup V_3$. Note that if these elements did not exist one set would be a subset of another or we would have an element in exactly two subsets. Then, to be closed, $u + v \in V_3$ since by the Exercise 12 lemma we cannot have $u + v$ be in either V_1 or V_2. Consider a, b in $\mathbb{F}$ where $a, b \neq 0, a - b \neq 0$. We also have $au + v \in V_3$ and $bu + v \in V_3$ since scalar multiplication by a non-zero constant should not change which subspaces a vector is in, so $au + v - (bu - v) = (a - b)u \in V_3$. However, u was explicitly only in V_1, so we have a contradiction and $V_1 \cup V_2 \cup V_3$ is not actually a subspace! Note that this proof does not work in $\mathbb{F}_2$ because we do not have 2 different numbers that satisfy the properties of a and b. □

15 Suppose U is a subspace of V. What is $U + U$?

Proof. We use the definition of subspace addition.

$$U + U = \{u + v : u, v \in U\}$$

Because U is a subspace and is therefore closed, this is identical to U. □

17 Is the operation of addition on the subspaces of V associative? In other words, if V_1, V_2, V_3 are subspaces of V, is

$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

Proof. Using the definition of subspace, we have

$$\begin{aligned}(V_1 + V_2) + V_3 &= \{(v_1 + v_2) + v_3 : v_1 \in V_1, v_2 \in V_2, v_3 \in V_3\} \\ &= \{v_1 + (v_2 + v_3) v_1 \in V_1, v_2 \in V_2, v_3 \in V_3\} \\ &= V_1 + (V_2 + V_3)\end{aligned}$$

using the associative property of vector addition. □

19 Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V_1 + U = V_2 + U$$

then $V_1 = V_2$

Proof. This is a nice application of the idea that subspaces do not have inverses. For example, we could have V as $\mathbb{R}^2$, $V_1 = \{(x, 0) : x \in \mathbb{R}\}$, and $V_1 = \{(0, y) : y \in \mathbb{R}\}$, and $U = V$. These both add to V, but they are not equal. □

21 Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbf{F}^5 : x, y \in F\}$$

Find a subspace W of $\mathbf{F}^5$ such that $\mathbf{F}^5 = U \oplus W$.

Proof. We have 2 variables, so we should have 3 variables in the other one to counteract this. The first 2 terms of U are just x and y, which is great because now they cannot vary. We can choose

$$W = \{(0, 0, p, q, r) \in \mathbf{F}^5 : p, q, r \in \mathbf{F}\}$$

And then any vector of the form (a, b, c, d, e) using

$$\begin{aligned}x &= a \\ y &= b \\ p &= c - (x + y) = c - a - b \\ q &= d - (x - y) = c - a + b \\ r &= e - (2x) = e - 2a\end{aligned}$$

□

23 Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V = V_1 \oplus U \text{ and } V = V_2 \oplus U$$

then $V_1 = V_2$.

Proof. This is untrue. Choose $V = \mathbb{R}^2$. Let $U = \{(x, 0) : x \in \mathbb{R}\}$. Then choose $V_1 = \{(0, b) : b \in \mathbb{R}\}$ but $V_2 = \{(a, a) : a \in \mathbb{R}\}$.

These are different subspaces, but it can be shown that their direct sum with U is both V. □

24 A function $f : \mathbb{R} \to \mathbb{R}$ is called *even* if

$$f(-x) = f(x)$$

for all $x \in \mathbb{R}$. A function $f : \mathbb{R} \to \mathbb{R}$ is called *odd* if

$$f(-x) = -f(x)$$

for all $x \in \mathbb{R}$. Let V_e denote the set of real-valued even functions on $\mathbb{R}$ and let V_o denote the set of real-valued odd functions on $\mathbb{R}$. Show that

$$\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$$

Proof. Given some function $f : \mathbb{R} \to \mathbb{R}$ we can construct it by adding the following functions:

We have $g \in V_e$,

$$g(x) = \frac{f(x) + f(-x)}{2}$$

It is clearly even by symmetry.

We can then choose $h \in V_o$,

$$h(x) = \frac{f(x) - f(-x)}{2}$$

This is clearly odd.

Now we must show that these are the *only* two functions that create f. Given some g and h, we must have $f(x) - g(x) = h(x)$. This means that

$$\begin{aligned}
f(-x) - g(-x) &= -h(x) \\
f(-x) - g(x) &= -h(x) \\
f(-x) &= g(x) - h(x) \\
f(x) &= g(x) + h(x) \\
f(-x) + f(x) &= 2g(x) \\
g(x) &= \frac{f(-x) + f(x)}{2}
\end{aligned}$$

h comes naturally afterward.

$\square$

2 Finite Dimensional Vector Spaces

We say goodbye to our infinite dimensional vector spaces (noooo not $\mathbb{R}^{\mathbb{R}}$) and we now focus on vector spaces which can actually be spanned by a list of vectors. This unit is... harder than the first one definitely, and it feels more like "linear algebra" than just "abstract algebra."

2.A Span and Linear Dependence

Not too bad.

1 Find a list of four distinct vectors in $\mathbf{F}^3$ whose span equals

$$\{(x, y, z) \in \mathbf{F}^3 : x + y + z = 0\}$$

Proof. Actually, I only need 2 vectors for this.

$$v_1 = (1, -1, 0) \text{ and } v_2 = (0, 1, -1)$$

With these, I can construct any vector that satisfies the properties. I can include some other ones but I am lazy.

□

3 Suppose $v_1, \ldots, v_m$ is a list of vectors in V. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k$$

Show that $\text{span}(v_1, \ldots, v_m) = \text{span}(w_1, \ldots, w_m)$.

Proof. Given an arbitrary linear combination of w_k,

$$\sum_{k=1}^{m} a_k w_k = \sum_{k=1}^{m} (\sum_{i=1}^{k} a_i) v_k$$

For a given v_k, we can construct it from a linear combination of w as $v_k = w_k - w_{k-1}$ where $v_1 = w_1$. Thus, since we can go both ways, the spans are equal.

□

5 Find a number t such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in $\mathbf{R}^3$

Proof. If we can find some

$$a(3,1,4) + b(2,-3,5) = (5,9,t)$$

then our list of vectors in not linearly independent. This turns into these equations:

$$\begin{aligned} 3a + 2b &= 5 \\ a - 3b &= 9 \\ 4a + 5b &= t \end{aligned}$$

We solve, and we get that $a = 3, b = -2, t = 2$, so it is not linearly independent when $\boxed{t = 2}$. □

7 (a) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$, then the list $1 + i, 1 - i$ is linearly independent.

Proof. Clearly there is no positive multiple of $1+i$ that creates $1-i$ you dumbass. The real and imaginary parts will always have the same sign. □

(b) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{C}$, then the list $1 + i, 1 - i$ is linearly dependent.

Proof. We can multiple $1 + i$ by

$$\begin{aligned} \frac{1-i}{1+i} &= \frac{(1-i)(1-i)}{(1+i)(1-i)} \\ &= \frac{-2i}{2} \\ &= -i \end{aligned}$$

to get $1 - i$, therefore its linearly dependent. □

9 Prove or give a counterexample: If $v_1, v_2, \dots, v_m$ is a linearly independent list of vectors in V, then

$$5v_1 - 4v_2, v_2, v_3, \dots, v_m$$

is linearly independent.

Proof. Suppose that $5v_1 - 4v_2, v_2, v_3, \dots, v_m$ was linearly dependent. Then,

$$a_1(5v_1 - 4v_2) + a_2v_2 + \cdots a_mv_m = 5a_1v_1 + (a_2 - 4a_1)v_2 + \cdots + a_mv_m = 0$$

and we have a contradiction. □

11 Prove or give a counterexample: If $v_1, \dots, v_m$ and $w_1, \dots w_m$ are linearly independent lists of vectors in V, then the list $v_1 + w_1, \dots, v_m + w_m$ is linearly independent.

Proof. We can construct a counterexample. Let us use $\mathbb{R}^1$ here. 3 is linearly independent, and so is -3, but $3 - 3 = 0$, which is not linearly independent. $\square$

13 Suppose $v_1, \dots v_m$ is linearly independent in V and $w \in V$. Show that

$v_1, \dots v_m$, w is linearly independent $\iff$ $w \notin \text{span}(v_1, \dots, v_m)$.

Proof. We prove the contrapositive in both directions.

Suppose that $w \in \text{span}(v_1, \dots, v_m)$. Then, there is some linear combination of $v_1, \dots, v_m$ and so $v_1, \dots, v_m, w$ is linearly dependent.

Suppose that $v_1, \dots, v_m, w$ is linearly dependent. Then, by the linear dependence lemma, there exists some element in this list that is in the span of the previous ones. Because all lists $v_1, \dots v_k, k \leq m$ are linearly independent, $w \in \text{span}(v_1, \dots, v_m)$. $\square$

15 Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbf{F})$.

Proof. We have a list of 5 polynomials that spans $\mathcal{P}_4(\mathbf{F})$. The length of a linearly independent list must be less than or equal to the length of this spanning list, so there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbf{F})$. $\square$

17 Prove that V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \dots$ of vectors in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m.

Proof. Because the length of the span of a vector space must be greater or equal to any given list of linearly independent vectors (Replacement theorem), this sequence means that there is no finite list of vectors that spans V.

If V is infinite-dimensional, we can construct such a sequence $v_1, v_2, \dots$. Begin by choosing an arbitrary vector $v_1 \neq 0$. Then, for each $k \geq 2$, we know that

$$\text{span}(v_1, \dots, v_{k-1}) \neq V$$

So there exists some $v_k \notin \text{span}(v_1, \dots, v_{k-1}$. Continue for an arbitrary number of steps. $\square$

19 Prove that the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.

Proof. I can construct a sequence of functions that are all linearly independent. In fact, I can use

$$1, x, x^2, x^3 \ldots$$

Since each polynomial is uniquely determined by its coefficients, these are linearly independent, and they are also clearly continuous. $\square$

20 Suppose $p_0, p_1, \ldots, p_m$ are polynomials in $\mathcal{P}_m(\mathbf{F})$ such that $p_k(2) = 0$ for each $k \in \{0, \ldots, m\}$. Prove that $p_0, p_1, \ldots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbf{F})$.

Proof. For $m = 0$, $p_0 = 0$, which isn't linearly independent.

For $m > 0$, if $p_k = 0$ for some $k \in \{0, \ldots, m\}$, then our set is already not linearly independent.

Suppose $p_k \neq 0$ for all $k \in \{0, \ldots, m\}$. Then, because $p_k(2) = 0$, all p_k has a factor of $(z - 2)$. When finding a linear combination that equals zero, we can factor out all by $(z - 2)$. That is

$$a_1 p_1 + a_2 p_2 + \cdots + a_m p_m = 0 \iff (z-2)(a_1 \frac{p_1}{z-2} + \cdots + a_m \frac{p_m}{z-2}) = 0$$

Now, we have a set of polynomials of degree $m - 1$, but we have $m + 1$ terms. For $\mathcal{F}_{m-1}(\mathbf{F})$ there is a list of m vectors that span it. Thus, the length of a linearly independent list of vectors must be less than or equal to m. We have $m + 1$ terms in our list, so we can determine that our list is linearly dependent. $\square$